Triton and the Cold Orb: inputs (blue = change me)

Quantity	Value	Unit	Source / note
Gravitational constant G	6.674E-11	$m^3\ kg^{-1}\ s^{-2}$	CODATA
Neptune mass	1.024E+26	kg	NASA fact sheet
Neptune radius	2.476E+07	m	equatorial, 1 bar
Neptune mean density	1638	kg/m^3	NASA fact sheet
Triton mass	2.139E+22	kg	Voyager 2
Triton radius	1.353E+06	m	Voyager 2
Triton mean density	2061	kg/m^3	Voyager 2
Triton orbit radius a_0	3.548E+08	m	semi-major axis
Triton orbital period	5.877	days	retrograde, inclination 157°
Neptune Love number k_2	0.41		published estimate
Seconds per year	31560000	s	
Triton surface temperature	38	K	Voyager 2 (1989)

Approximate model: constant k_2/Q, circular orbit, inclination ignored. Neptune's Q is the uncertain input (see 'Tidal decay').

The perch and the tap: Roche limits and the tidal ratio

Quantity	Value	Unit	Formula / reading
Orbit radius in Neptune radii	14.33	R_N	a_o / R_N
Orbital speed	4.39	km/s	$2\pi a_o$ / P
Roche limit, rigid	28,901	km	R_N (2ρ_N/ρ_T)^(1/3)
Roche limit, rigid	1.17	R_N	
Roche limit, fluid	55,970	km	2.44 R_N (ρ_N/ρ_T)^(1/3)
Roche limit, fluid	2.26	R_N	
Tidal stretch ÷ self-gravity, today	0.00053	ratio	$2GM_N R_T/a^3 \div Gm_T/R_T^2$: the perch
Safety margin today	1,881	×	how many times stronger self-gravity is
Tidal ratio at rigid Roche limit	0.98	ratio	≈ 1: the tap, where the orb breaks
Tidal ratio at fluid Roche limit	0.14	ratio	

Drag: how fast Triton falls inward (retrograde tides)

$da/dt = -3\,(k_2/Q)(m/M)(R_N/a)^5\, n\, a \;\Rightarrow\; t = (2/13)(a_0\text{^}6.5 - a_R\text{^}6.5) / K,\; K = 3(k_2/Q)(m/M)\, R_N^5\, \sqrt{(GM)}$

Neptune Q (change me)	K (m^6.5/s)	Inward drift today (mm/yr)	Years to fluid Roche limit (Gyr)	Years to rigid Roche limit (Gyr)
9000	2.198E+37	6.554	8.3	8.3
30000	6.593E+36	1.966	27.8	27.8
100000	1.978E+36	0.590	92.5	92.5
330000	5.994E+35	0.179	305.3	305.3

HAT note: Neptune's Q is poorly known (≈ 9,000 to 330,000). The direction is certain (inward); the time is uncertain by more than 10×. The often-quoted ≈ 3.6 Gyr (Chyba et al., 1989) assumes stronger dissipation than the lowest case here.

Triton as a scale analog of the Cold Orb (Metaphor; numbers real)

Cold Orb idea	Triton	Number	Shelf
Cold	Surface ≈ 38 K, nitrogen ice	38 K	Established
Perch	Held at 14 Neptune radii, self-gravity far stronger than tide	1,881× margin	Derived
Retro orbit	Circles Neptune backwards (margin note: 'RETRO orbits')	inclination 157°	Established
Drag	Retrograde tides brake it; it falls inward	6.6 mm/yr (Q = 9,000)	Derived
Tap	Tide equals self-gravity at the Roche limit	0.98 at rigid limit	Derived
Ring formation	Broken Triton likely becomes a ring	billions of years	Candidate (published)
Cold but alive	Nitrogen geysers about 8 km high; possible ocean	Voyager 2, 1989	Established
Captured	Likely a Kuiper Belt object captured from a binary pair	Agnor & Hamilton, 2006	Candidate (published)

The medium: 3D within 4D, bilateral, and the pluck

Point	Statement	Shelf	Credit
Bilateral motion	Spacetime tells matter how to move; matter tells spacetime how to curve.	Established	John Wheeler
Ripples	Gravitational waves measured in 2015 from a merger ≈ 1.3 billion light-years away.	Established	LIGO
Speed of the pluck	Gravity travels at c (GW170817, 2017: equal to c within about 1 part in 10^{15}). Not instantaneous.	Established	LIGO / Virgo / Fermi
Travels forever	Wave strength falls as 1/distance and never reaches zero.	Established	
String speed	A wave on a string moves at $\sqrt{(T/\mu)}$.	Established	Mersenne (1636)
Snow globe	The 3D orb is a snow globe in the hand of God.	Faith	Owner
Infinite field	An infinite universe has an infinite field.	Postulated (boundless certain; infinite Open)	Owner